

$$aA \rightarrow x_a A$$

$$bB \rightarrow x_b B$$

Final:-

$$S_0 \rightarrow AX_1 \mid BA \mid AA \mid AB \mid x_a A \mid x_b B \mid a \mid b \mid \epsilon$$

$$S \rightarrow AX_1 \mid BA \mid AA \mid AB \mid x_a A \mid x_b B \mid a \mid b \mid \epsilon$$

$$A \rightarrow x_a A \mid a$$

$$B \rightarrow x_b B \mid b$$

$$X_1 \rightarrow BA$$

$$x_a \rightarrow a$$

$$x_b \rightarrow b$$

(Ques 17) Is the following grammar ambiguous?

i) $E \rightarrow E + E \mid E * E \mid id$ ii) $E \rightarrow E + E \mid E * E \mid (E) \mid \epsilon$

→ A grammar is ambiguous if there exists at least one string that has more than one parse tree.

i) Grammar

$$E \rightarrow E + E \mid E * E \mid id$$

